

FIM & WAZUH AGENT

DEPLOYMENT ACROSS WINDOWS' LINUX' AND UBUNTU



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File Integrity Monitoring With MD5Sum

File Integrity Monitoring with MD5Sum involves calculating and comparing MD5 checksums to verify whether a file's content has changed. By checking these hashes, you can determine if a file has been tampered with, ensuring its integrity and security. This method is crucial for detecting unauthorized modifications, helping organizations protect sensitive data and respond swiftly to potential threats.

1st Step

Here we first make a directory on any name here I give fim.

```
(kali㉿kali)-[~]  
$ mkdir fim  
  
(kali㉿kali)-[~]  
$ ls  
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  fim  
  
(kali㉿kali)-[~]  
$ cd fim
```

2nd Step

Then I make 3 file through nano.

```
(kali㉿kali)-[~/fim]  
$ nano file1.txt  
  
(kali㉿kali)-[~/fim]  
$ nano file2.txt  
  
(kali㉿kali)-[~/fim]  
$ nano file3.txt  
  
(kali㉿kali)-[~/fim]  
$ ls  
file1.txt  file2.txt  file3.txt
```

3rd Step

Now we will use MD5SUM to extract the hashes of these 3 files which I created.

```
(kali@kali)-[~/fim]
$ sudo md5sum *.txt
[sudo] password for kali:
517984489e9f1e008caa696f01c3c251  file1.txt
db06069ef1c9f40986ffa06db4fe8fd7  file2.txt
7c28ecaa2c23703db0f79abe7c930f60  file3.txt
```

4th Step

Now we will save these hashes in checklist.chk

```
(kali@kali)-[~/fim]
$ sudo su
(root@kali)-[/home/kali/fim]
# md5sum *.txt > checklist.chk

(root@kali)-[/home/kali/fim]
# ls
checklist.chk  file1.txt  file2.txt  file3.txt
```

5th Step

Now we will check the hashes

```
(kali@kali)-[~/fim]
$ md5sum -c checklist.chk
file1.txt: OK
file2.txt: OK
file3.txt: OK

(kali@kali)-[~/fim]
```

6th Step

Now we will modify a file and check its hash again. After changing `file2` and rechecking its hash, it shows that `file2` has failed the integrity check because its hash has changed.

```
(kali㉿kali)-[~/fim]
$ nano file2.txt

(kali㉿kali)-[~/fim]
$ md5sum -c checklist.chk
file1.txt: OK
file2.txt: FAILED
file3.txt: OK
md5sum: WARNING: 1 computed checksum did NOT match
```

MAKING GROUPS FOR AGENTS

Wazuh Overview

Recently viewed

- Home
- Explore
- Endpoint security
- Threat intelligence
- Security operations
- Cloud security
- Server management**
 - Endpoints Summary
 - Endpoint Groups**
 - Rules
 - Decoders
 - CDB Lists
 - Status
 - Cluster

LAST 24 HOURS ALERTS

Critical severity	High severity	Medium severity	Low severity
0	0	76	2482
Rule level 15 or higher	Rule level 12 to 14	Rule level 7 to 11	Rule level 4 to 6

THREAT INTELLIGENCE

- Malware Detection: Verify that your systems are configured according to your security policies baseline.
- Threat Hunting: Browse through your security alerts, identifying issues and threats in your environment.
- Vulnerability Detection: Discover what applications in your environment are affected by well-known vulnerabilities.
- MITRE ATT&CK: Security events from the knowledge base of adversary tactics and techniques based on real-world observations.
- VirusTotal: Alerts resulting from VirusTotal analysis of suspicious files via integration with their API.

Click on add new group and give name. Here I added linux and window.

Wazuh Endpoint Groups

Groups (3)

From here you can list and check your groups, its agents and files.

Search

WQL

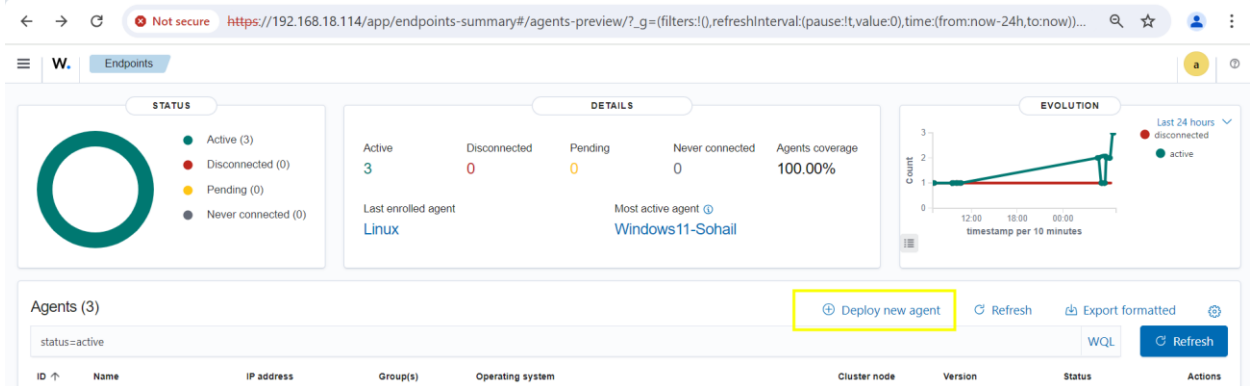
Actions

Name ↑	Agents	Configuration	Actions
default	0	...	...
Linux	2	...	...
Windows	1	...	...

Rows per page: 10

Deploying Agent in Window

1. Start Wazuh open it in your browser by its IP.
2. Click on Deploy new agent.



3. Click on Window MSI 32/64 bits

[https://192.168.18.114/app/endpoints-summary#/agents-preview/?_g=\(filters:!\(\),refreshInterval:\(pause:!t,value:0\),time:\(from:now-24h,to:now\)\)...](https://192.168.18.114/app/endpoints-summary#/agents-preview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))...)

✓ **Select the package to download and install on your system:**

LINUX

☐ RPM amd64 ☐ RPM aarch64

☐ DEB amd64 ☐ DEB aarch64

WINDOWS

☒ MSI 32/64 bits

macOS

☐ Intel

☐ Apple silicon

❗ For additional systems and architectures, please check our [documentation](#).

✓ **Server address:**

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address ?

☒ Remember server address



Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: [?](#)

Window11

[?](#) The agent name must be unique. It can't be changed once the agent has been enrolled. [?](#)

Select one or more existing groups: [?](#)

Windows X



4

Run the following commands to download and install the agent:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.8.1-1.msi -OutFile  
${env.tmp}\wazuh-agent; msixec.exe /i ${env.tmp}\wazuh-agent /q WAZUH_MANAGER='192.168.18.114'  
WAZUH_AGENT_GROUP='Windows' WAZUH_AGENT_NAME='Window11'
```

[?](#) Requirements

- You will need administrator privileges to perform this installation.
- PowerShell 3.0 or greater is required.

Keep in mind you need to run this command in a Windows PowerShell terminal.

5

Start the agent:

```
NET START WazuhSvc
```

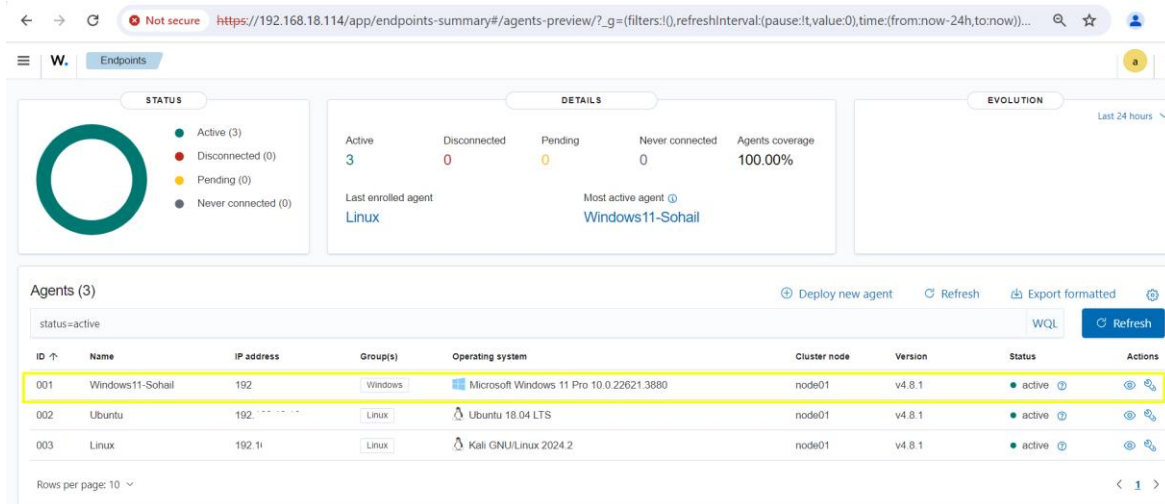
Run power shell as administrator and run these commands.

```
PS C:\Windows\system32> Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.8.1-1.msi -OutFile $(env:tmp)\wazuh-agent; msexec.exe /i $(env:tmp)\wazuh-agent /q WAZUH_MANAGER="192.168.18.145" WAZUH_AGENT_NAME="window"
PS C:\Windows\system32> NET START WazuhSvc

The Wazuh service was started successfully.

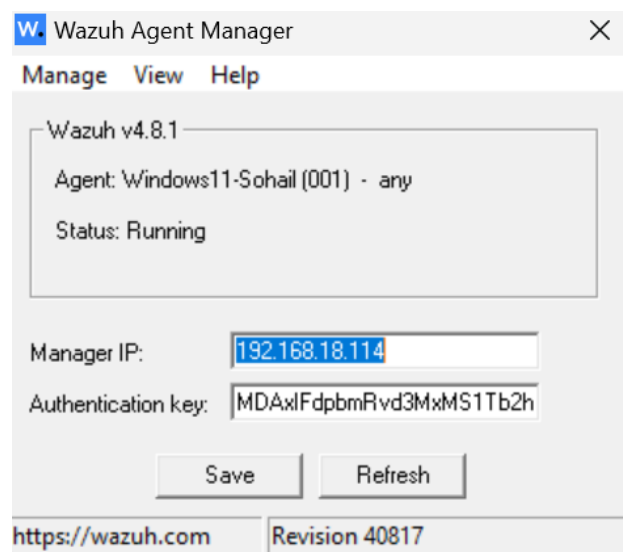
PS C:\Windows\system32>
```

Now we will check the wazuh



CONFIGURE FIM IN Window

1. Open the manage agent from administrator (from system start) and click on manage and stop it first.



2. Then click on view and view config and find File integrity monitoring section, and edit this now if we want to monitor a partition so we will add there complete location.

```
<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <!-- Default files to be monitored. -->
  <directories recursion_level="0" restrict="regedit.exe$|system.ini$|win.ini$" %WINDIR%</directories>

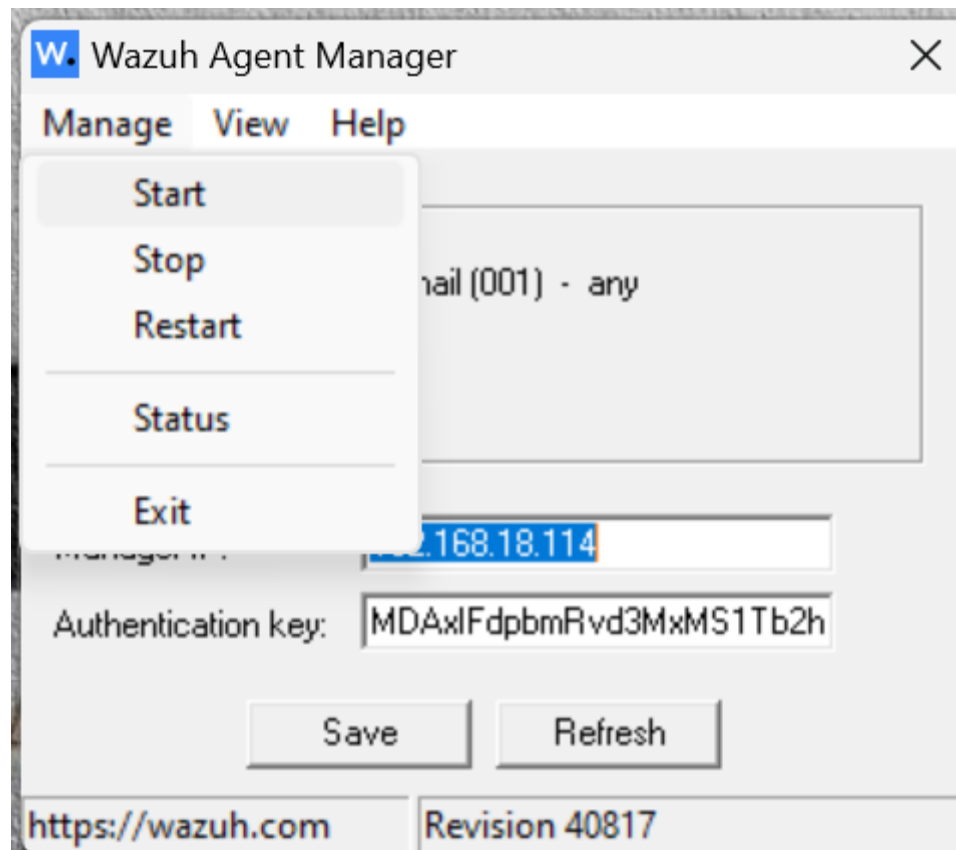
  <directories recursion_level="0" restrict="at.exe$|attrib.exe$|cacls.exe$|cmd.exe$|eventcreate.exe$|ftp.exe$|lsass.exe$|net.exe$|net1.exe$|netsh.exe$|reg.exe$|regedt32.exe$|regsvr32.exe$|runas.exe$|sc.exe$|schtasks.exe$|sethc.exe$|subst.exe$" %WINDIR%\SysNative</directories>
  <directories recursion_level="0" restrict="WMIC.exe$" %WINDIR%\SysNative\wbem</directories>
  <directories recursion_level="0" restrict="powershell.exe$" %WINDIR%\SysNative\WindowsPowerShell\v1.0</directories>
  <directories recursion_level="0" restrict="winrm.vbs$" %WINDIR%\SysNative</directories>

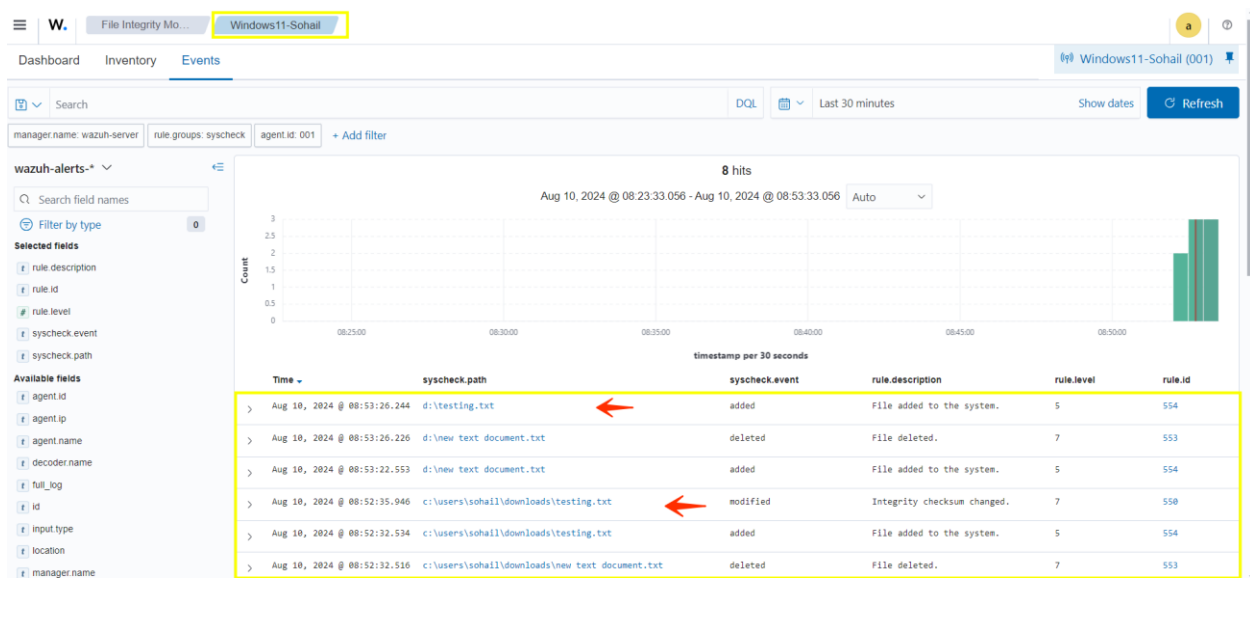
  <!-- 32-bit programs. -->
  <directories recursion_level="0" restrict="at.exe$|attrib.exe$|cacls.exe$|cmd.exe$|eventcreate.exe$|ftp.exe$|lsass.exe$|net.exe$|net1.exe$|netsh.exe$|reg.exe$|regedit.exe$|regedt32.exe$|regsvr32.exe$|runas.exe$|sc.exe$|schtasks.exe$|sethc.exe$|subst.exe$" %WINDIR%\System32</directories>
  <directories recursion_level="0" restrict="WMIC.exe$" %WINDIR%\System32\wbem</directories>
  <directories recursion_level="0" restrict="powershell.exe$" %WINDIR%\System32\WindowsPowerShell\v1.0</directories>
  <directories recursion_level="0" restrict="winrm.vbs$" %WINDIR%\System32</directories>

  <directories realtime="yes" %PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup</directories>
  <directories report_changes="yes" check_all="yes" realtime="yes" C:\Users\Sohail\Downloads</directories>
  <directories report_changes="yes" check_all="yes" realtime="yes" D:\</directories>

  <ignore>%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup\desktop.ini</ignore>
```

3. Now we will start the wazuh agent manager

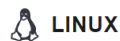




Deploying Agent Linux

re [https://192.168.18.114/app/endpoints-summary#/agents-preview/?_g=\(filters:!,refreshInterval:\(pause:!t,value:0\),time:\(from:now-24h,to:now\)\)](https://192.168.18.114/app/endpoints-summary#/agents-preview/?_g=(filters:!,refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now)))

Select the package to download and install on your system.



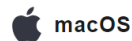
LINUX

- ☐ RPM amd64 ☐ RPM aarch64
- ☒ DEB amd64 ☐ DEB aarch64



WINDOWS

- ☐ MSI 32/64 bits



macOS

- ☐ Intel
- ☐ Apple silicon

For additional systems and architectures, please check our [documentation](#).

Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address ?

192.168.18.114



Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: [?](#)

Linux

[?](#) The agent name must be unique. It can't be changed once the agent has been enrolled. [↗](#)

Select one or more existing groups: [?](#)

Linux X



4

Run the following commands to download and install the agent:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.1-1_amd64.deb && sudo
WAZUH_MANAGER='192.168.18.114' WAZUH_AGENT_GROUP='Linux' WAZUH_AGENT_NAME='Linux' dpkg -i ./wazuh-
agent_4.8.1-1_amd64.deb
```

[?](#) Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

5

Start the agent:

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

Close

Now we will run these commands in linux terminal.

```
(kali@kali)-[~]
$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.1-1_amd64.deb && sudo WAZUH_MANAGER='192.168.18.114' WAZUH_AGENT_GRO
P='Linux' WAZUH_AGENT_NAME='Linux' dpkg -i ./wazuh-agent_4.8.1-1_amd64.deb
--2024-08-10 18:21:10-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.1-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com) ... 2600:9000:20c7:f000:8:fed3:b0c0:93a1, 2600:9000:20c7:3000:8:fed3:b0c0:93a1, ...
Connecting to packages.wazuh.com (packages.wazuh.com)[2600:9000:20c7:f000:8:fed3:b0c0:93a1]:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10270680 (9.8M) [binary/octet-stream]
Saving to: 'wazuh-agent_4.8.1-1_amd64.deb'

wazuh-agent_4.8.1-1_amd64.deb 100%[=====] 9.79M 2.41MB/s in 4.4s

2024-08-10 18:21:15 (2.25 MB/s) - 'wazuh-agent_4.8.1-1_amd64.deb' saved [10270680/10270680]

[sudo] password for kali:
Selecting previously unselected package wazuh-agent.
(Reading database ... 390905 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.8.1-1_amd64.deb ...
Unpacking wazuh-agent (4.8.1-1) ...
Setting up wazuh-agent (4.8.1-1) ...

(kali@kali)-[~]
$ sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /usr/lib/systemd/system/wazuh-agent.service.
```

Now Check the wazuh which is open in browser and refresh it

← → ↻ 🔒 Not secure https://192.168.18.114/app/endpoints-summary#/agents-preview/?g=(filters:[]refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now))... 🔍 ☆ 👤 ⋮

W. Endpoints

STATUS

- Active (3)
- Disconnected (0)
- Pending (0)
- Never connected (0)

DETAILS

Active	Disconnected	Pending	Never connected	Agents coverage
3	0	0	0	100.00%

Last enrolled agent: Linux

Most active agent: Windows11-Sohail

EVOLUTION

Last 24 hours

Agents (3)

Deploy new agent Refresh Export formatted Refresh

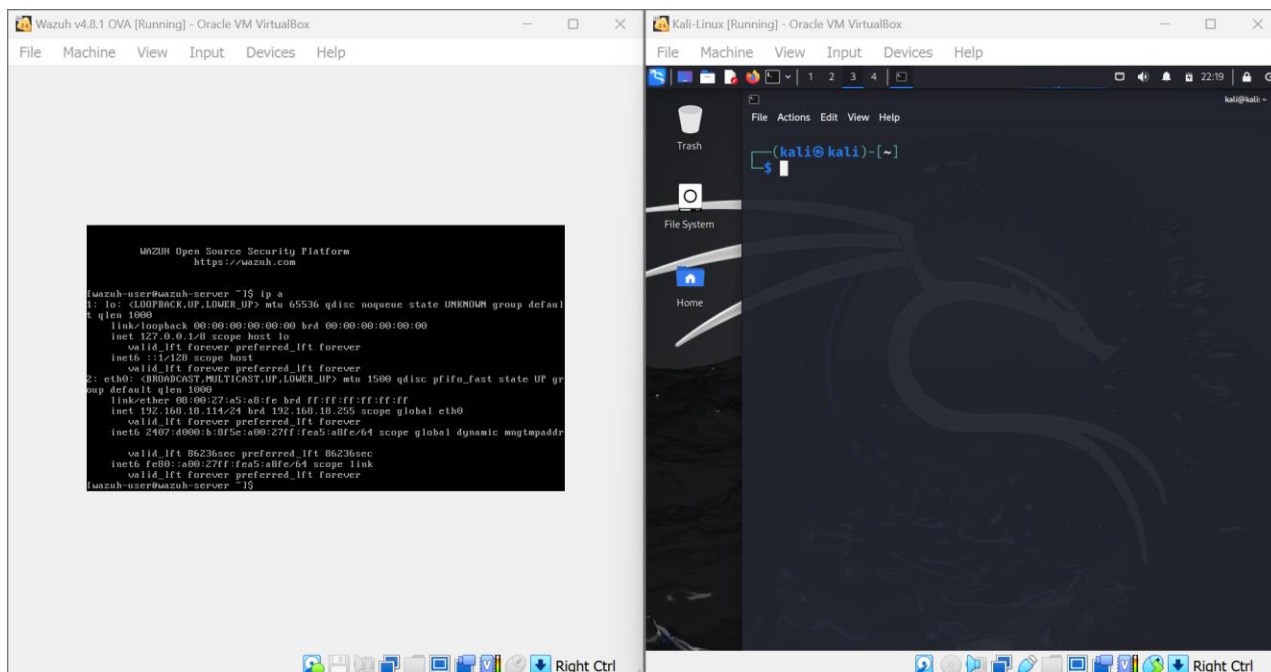
Search WQL

ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	Windows11-Sohail	192.168.18.143	Windows	Microsoft Windows 11 Pro 10.0.22621.3880	node01	v4.8.1	active	
002	Ubuntu	192.168.18.13	Linux	Ubuntu 18.04 LTS	node01	v4.8.1	active	
003	Linux	192.168.18.146	Linux	Kali GNU/Linux 2024.2	node01	v4.8.1	active	

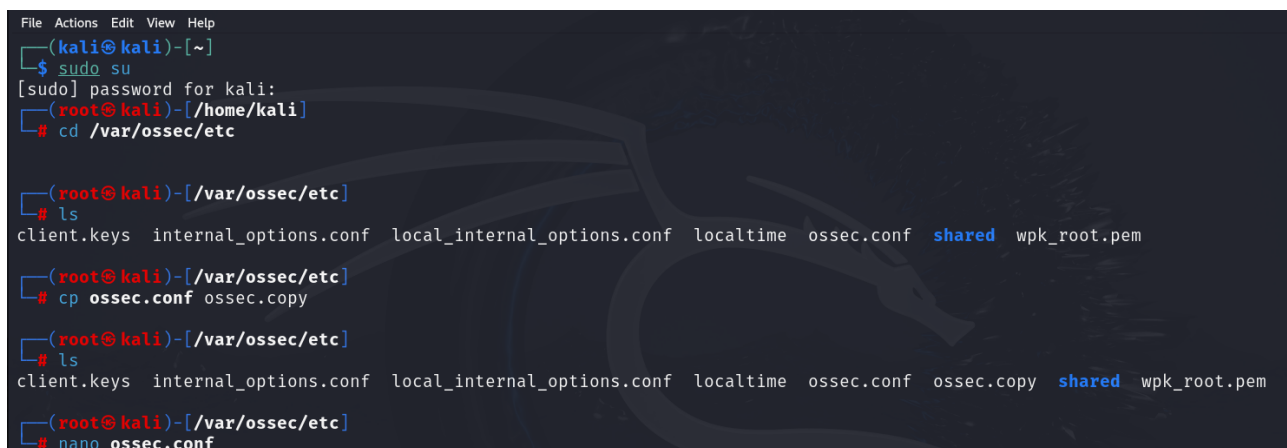
Rows per page: 10

CONFIGURE FIM IN Linux

My wazuh and linux are running



First we switch to root and then go to /var/ossec/etc



Nano the file of ossec.conf and add the directory which we want to monitor.

```

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <scan_on_start>yes</scan_on_start>

  <!-- Directories to check (perform all possible verifications) -->
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>
  <directories check_all="yes" whodata="yes">/home/kali/Downloads</directories>

```

Here I add the Downloads path, to get exact this path

```

File  Actions  Edit  View  Help
(kaliⓈkali)-[~]
$ cd Downloads

(kaliⓈkali)-[~/Downloads]
$ pwd
/home/kali/Downloads

(kaliⓈkali)-[~/Downloads]
$

```

- Then run this command:
`sudo systemctl restart wazuh-agent.`
- Now we edit the file or remove.

```

(kali㉿kali)-[~/Downloads]
$ sudo rm testingFim1.txt

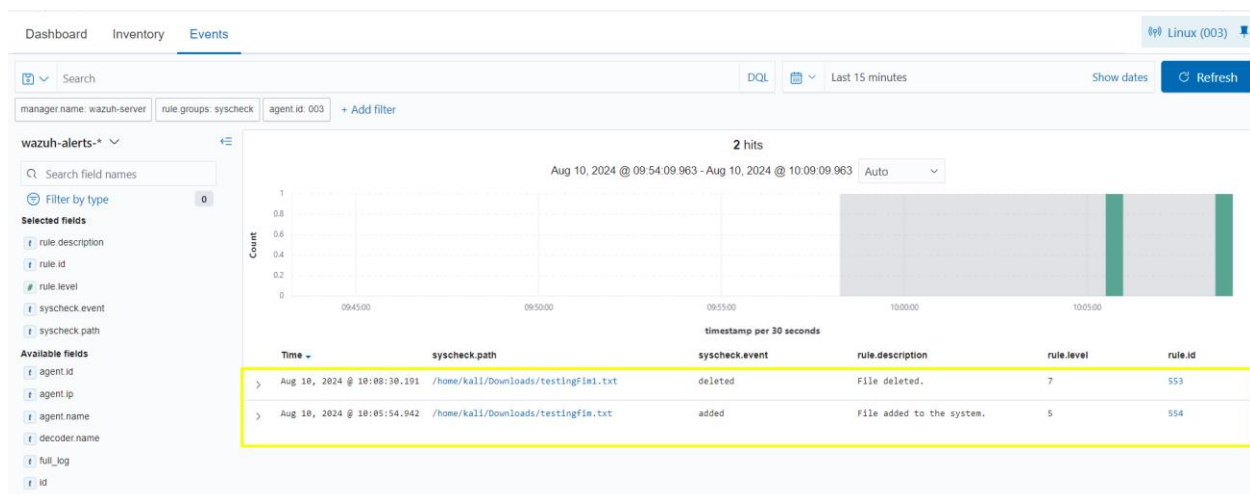
(kali㉿kali)-[~/Downloads]
$ LS
LS: command not found

(kali㉿kali)-[~/Downloads]
$ ls
testingFim.txt  testingfim.txt

```

- Now we will refresh the wazuh page to check that the logs are received or not.

Here the logs are received.



Now I will open add file which is showing in the above picture.

/home/kali/Downloads/testingFim.txt

Details

 Last analysis Aug 10, 2024 @ 10:04:46.000	 Last modified Aug 10, 2024 @ 09:48:44.000	 User root
 User ID 0	 Group root	 Group ID 0
 Size 20 Bytes	 Inode 1048677	 MD5 e86a04d8c01ad39ab72f5a4a426111df
 SHA1 d8505269de4842259912edc17565a5cfe757df19	 SHA256 713a00fa62e7b125a86942f74f65b9cacad78176466a31e8f5f871a60752d60c	
 Permissions rw-r--r--		

Recent events

0 hits

Search	DQL	 Last 15 minutes	Show dates	 Refresh
+ Add filter				

ADDING AGENT UBUNTU

- First on the wazuh manager and then copy their IP and run in browser to on the Wazuh then add new agent.

✓ Select the package to download and install on your system:

 **LINUX**
☐ RPM amd64 ☐ RPM aarch64
☒ DEB amd64 ☐ DEB aarch64

 **WINDOWS**
☐ MSI 32/64 bits

 **macOS**
☐ Intel
☐ Apple silicon

❗ For additional systems and architectures, please check our [documentation](#).

✓ Server address:

This is the address the agent uses to communicate with the server. Enter an IP address or a fully qualified domain name (FQDN).

Assign a server address ?

192.168.18.114

wazuh server IP

☒ Remember server address

✓ Optional settings:

By default, the deployment uses the hostname as the agent name. Optionally, you can use a different agent name in the field below.

Assign an agent name: ?

Ubuntu

❗ The agent name must be unique. It can't be changed once the agent has been enrolled.

Select one or more existing groups: ?

Linux ×

4

Run the following commands to download and install the agent:

```
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.1-1_amd64.deb && sudo  
WAZUH_MANAGER='192.168.18.114' WAZUH_AGENT_GROUP='Linux' WAZUH_AGENT_NAME='Ubuntu' dpkg -i ./wazuh-  
agent_4.8.1-1_amd64.deb
```

④ Requirements

- You will need administrator privileges to perform this installation.
- Shell Bash is required.

Keep in mind you need to run this command in a Shell Bash terminal.

5

Start the agent:

```
sudo systemctl daemon-reload  
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent
```

Close

- Run these commands in ubuntu terminal

```
@ubuntu:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.1-1_amd64.deb && sudo WAZUH_M
ANAGER='192.168.18.145' WAZUH_AGENT_GROUP='Linux-os' WAZUH_AGENT_NAME='ubuntu' dpkg -i ./wazuh-agent_4.8.1-1_amd64.deb
--2024-08-10 21:31:00-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.8.1-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 13.35.18.20, 13.35.18.74, 13.35.18.103, ...
Connecting to packages.wazuh.com (packages.wazuh.com)[13.35.18.20]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10270680 (9.8M) [binary/octet-stream]
Saving to: 'wazuh-agent_4.8.1-1_amd64.deb'

wazuh-agent_4.8.1-1 100%[=====] 9.79M 2.47MB/s in 4.2s

2024-08-10 21:31:06 (2.34 MB/s) - 'wazuh-agent_4.8.1-1_amd64.deb' saved [10270680/10270680]

[sudo] password for fox:
Selecting previously unselected package wazuh-agent.
(Reading database ... 206466 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.8.1-1_amd64.deb ...
Unpacking wazuh-agent (4.8.1-1) ...
Setting up wazuh-agent (4.8.1-1) ...
@ubuntu:~$ sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /lib/systemd/system/wazuh-agent.service.
```

- Now we will check the wazuh and refresh they will show that the agent are deployed.

W. Endpoints

a

STATUS

- Active (3)
- Disconnected (0)
- Pending (0)
- Never connected (0)

DETAILS

Active: 3

Disconnected: 0

Pending: 0

Never connected: 0

Agents coverage: 100.00%

Last enrolled agent: Linux

Most active agent: Windows11-Sohail

EVOLUTION

Last 30 minutes

Agents (3)

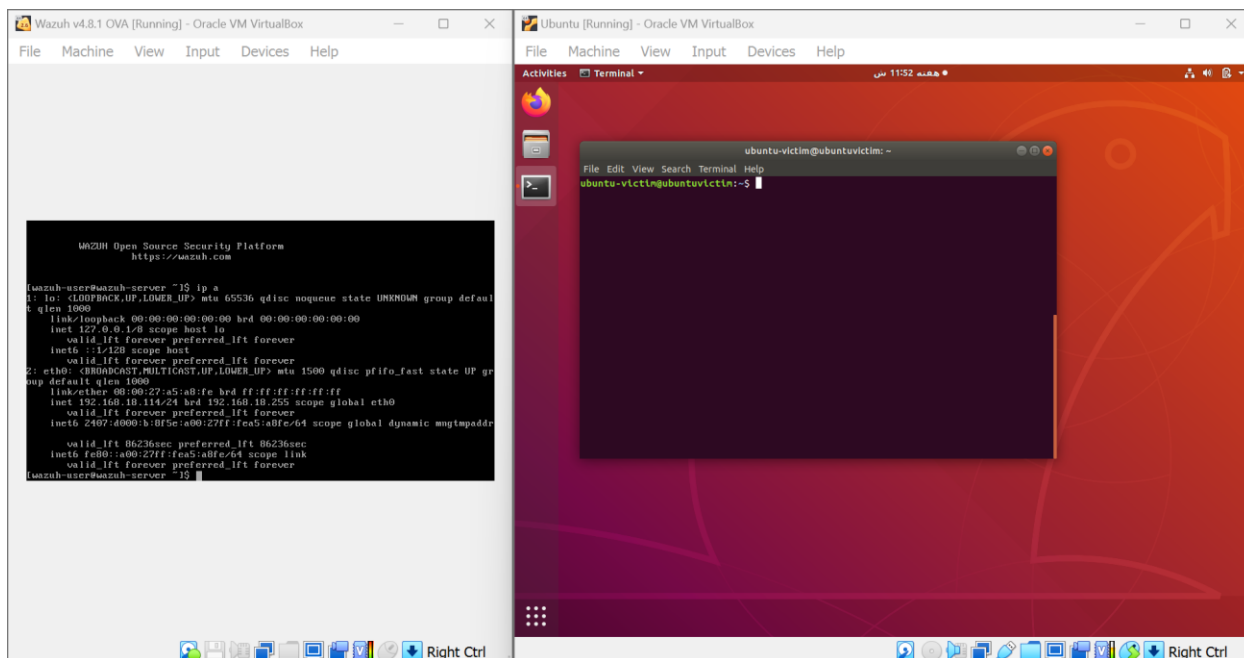
[Deploy new agent](#)
[Refresh](#)
[Export formatted](#)

[WQL](#)
[Refresh](#)

ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
001	Windows11-Sohail	192.168.18.13	Windows	Microsoft Windows 11 Pro 10.0.22621.3880	node01	v4.8.1	active	View Refresh
002	Ubuntu	192.168.18.13	Linux	Ubuntu 18.04 LTS	node01	v4.8.1	active	View Refresh
003	Linux	192.168.18.146	Linux	Kali GNU/Linux 2024.2	node01	v4.8.1	active	View Refresh

CONFIRGURE FIM IN UBUNTU

1. My wazuh and ubuntu are running.



2. To configure ubuntu in FIM machine we must be in root first

Sudo -i

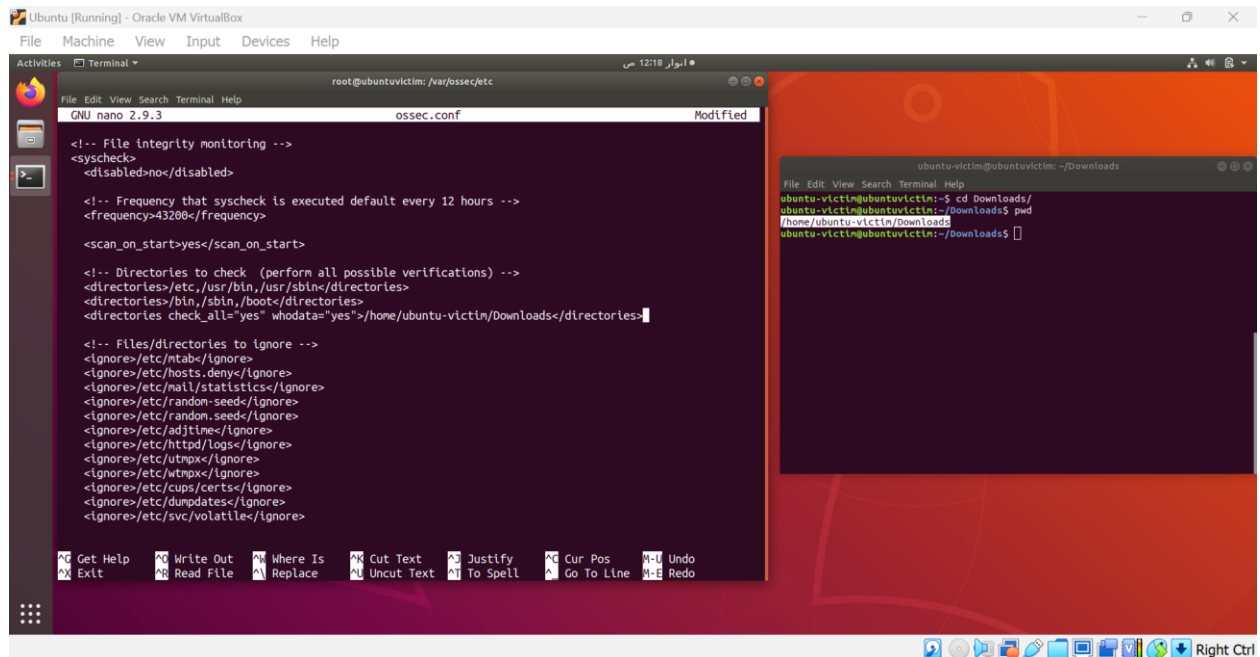
```
ubuntu-victim@ubuntu-victim:~$ sudo -i
[sudo] password for ubuntu-victim:
```

3. Now open the ossec.cnf file

```
ubuntu-victim@ubuntu-victim:~$ sudo -i
[sudo] password for ubuntu-victim:
root@ubuntu-victim:~# cd /var/ossec/etc
root@ubuntu-victim:/var/ossec/etc# ls
client.keys  internal_options.conf  local_internal_options.conf  localtime  ossec.conf  shared  wpk_root.pem
root@ubuntu-victim:/var/ossec/etc# nano ossec.conf
```

4. Path are added.

After adding this then ctrl+o then Enter and then ctrl+x



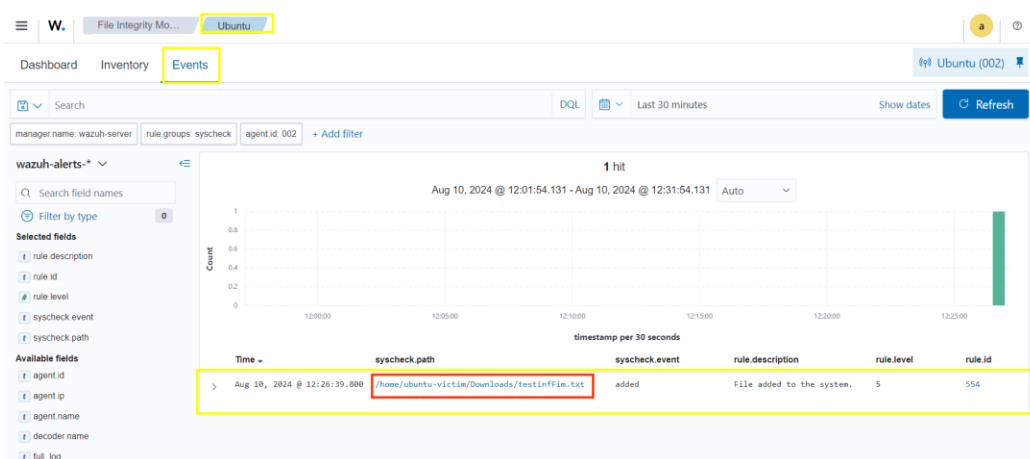
5. Restart wazuh agent now

```
root@ubuntuvictim:/var/ossec/etc# nano ossec.conf
root@ubuntuvictim:/var/ossec/etc# systemctl restart wazuh-agent
```

6. Here I add testinfim.txt file

```
ubuntu-victim@ubuntuvictim:~/Downloads$ sudo nano testinfim.txt
[sudo] password for ubuntu-victim:
ubuntu-victim@ubuntuvictim:~/Downloads$ ls
juice-shop-14.5.1_node16_linux_x64.tgz  slideshow-gallery.1.4.6  wordpress-5.0
mail-masta.zip                          slideshow-gallery.1.4.6.zip  wordpress-5.0.zip
reflex-gallery.3.1.3.zip                 testinfim.txt
```

7. Now I will check it in wazuh



8. Now I will click on this to check more details.

timestamp per 30 seconds

Time	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Aug 10, 2024 @ 12:26:39.800	/home/ubuntu-victim/Downloads/testinffim.txt	added	File added to the system.	5	554

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Table

JSON

† _index	wazuh-alerts-4.x-2024.08.10
† agent.id	002
† agent.ip	192.168.18.13
† agent.name	Ubuntu
† decoder.name	syscheck_new_entry
† full_log	File '/home/ubuntu-victim/Downloads/testinffim.txt' added Mode: realtime
† id	1723317999.1571662707
† input.type	log
† location	syscheck
† manager.name	wazuh-server
† rule.description	File added to the system.
# rule.firedtimes	1
† rule.gdpr	II_5.1.f